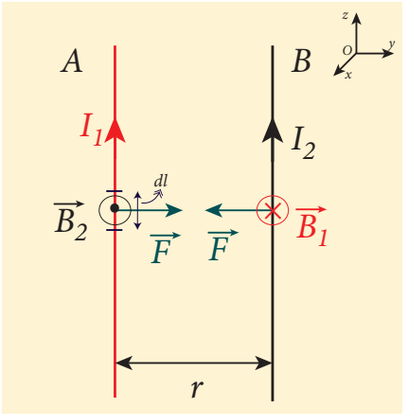


Figure 3.54 Current in both the conductors are in the same direction - attracts each other

The force acting per unit length of the conductor A due to the current in conductor B is

$$\frac{\vec{F}}{l} = \frac{\mu_\circ I_1 I_2}{2\pi r}\hat{j}$$

Thus the force between two parallel current carrying conductors is attractive if they carry current in the same direction. (Figure 3.55)

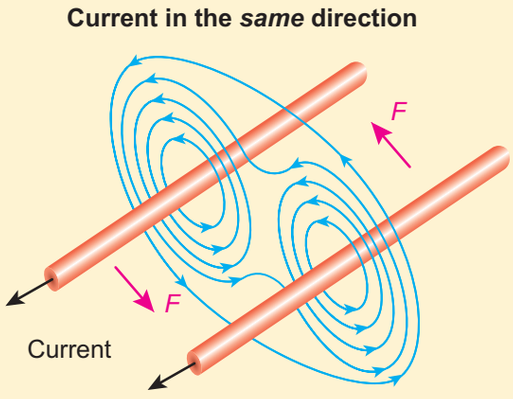


Figure 3.55 Two parallel conductors carrying current in same direction experience an attractive force

The force between two parallel current carrying conductors is repulsive if they carry current in opposite directions (Figure 3.56).

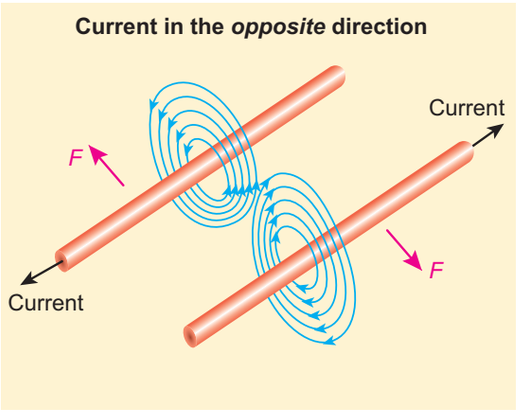


Figure 3.56 Two parallel conductors carrying current in opposite direction experience a repulsive force

Definition of ampere

One ampere is defined as that constant current which when passed through each of the two infinitely long parallel straight conductors kept side by side parallelly at a distance of one metre apart in air or vacuum causes each conductor to experience a force of 2×10^{-7} newton per metre length of conductor.

3.11 TORQUE ON A CURRENT LOOP

The force on a current carrying wire in a magnetic field is responsible for the motor operation.

3.11.1 Torque on a current loop placed in a magnetic field

Consider a rectangular loop PQRS carrying current I is placed in a uniform magnetic field $\vec{B}$. Let a and b be the length and breadth of rectangular loop respectively. The unit vector $\hat{n}$ normal to the plane of the

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